Cloud

Using Sentinel-2 images to identify clouds in Hydrosat imagery



Robin Alvarez

Hydrosat

- Global leader in thermal imagery, operating from the U.S. and Luxembourg.
- Provides crucial information concerning farmers, governments, and other corporations worldwide.
- Two satellites: VanZyl-01 and VanZyl-02



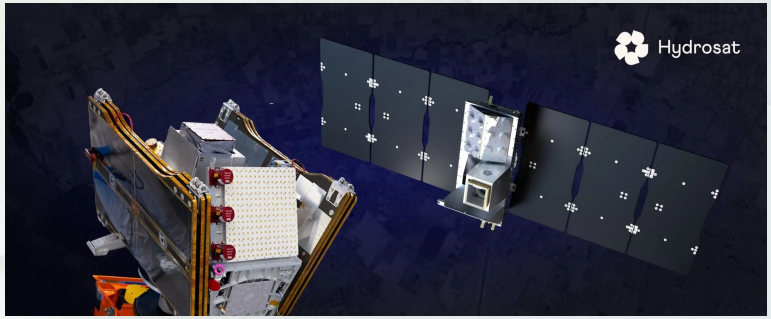
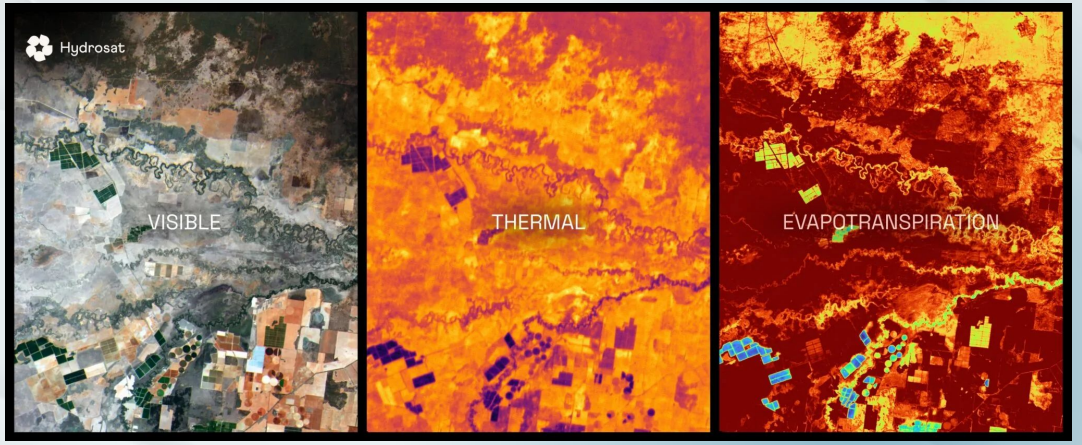


Image Credit: Hydrosat

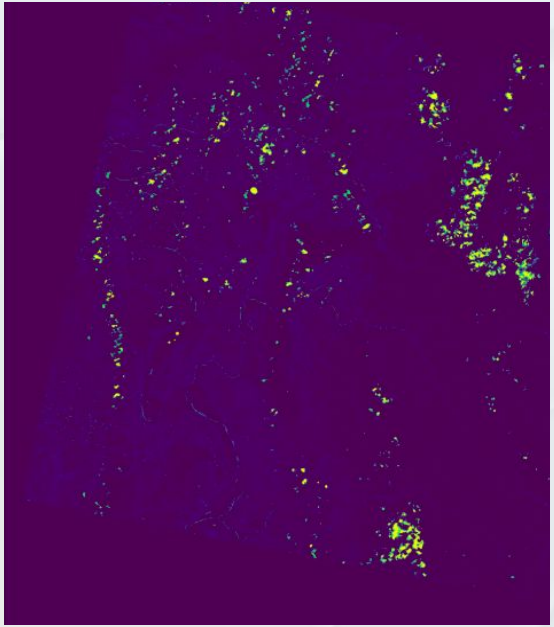


My Project: Creating a cloud mask

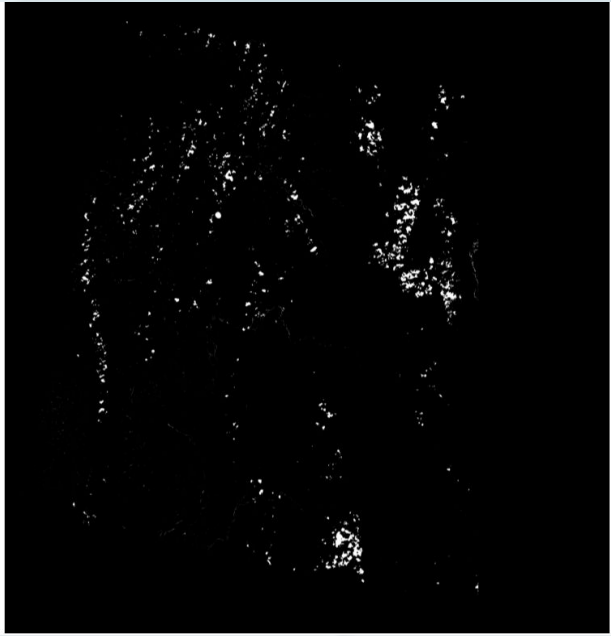


Original Image

Image Credit: ESA Sentinel-2, Copernicus

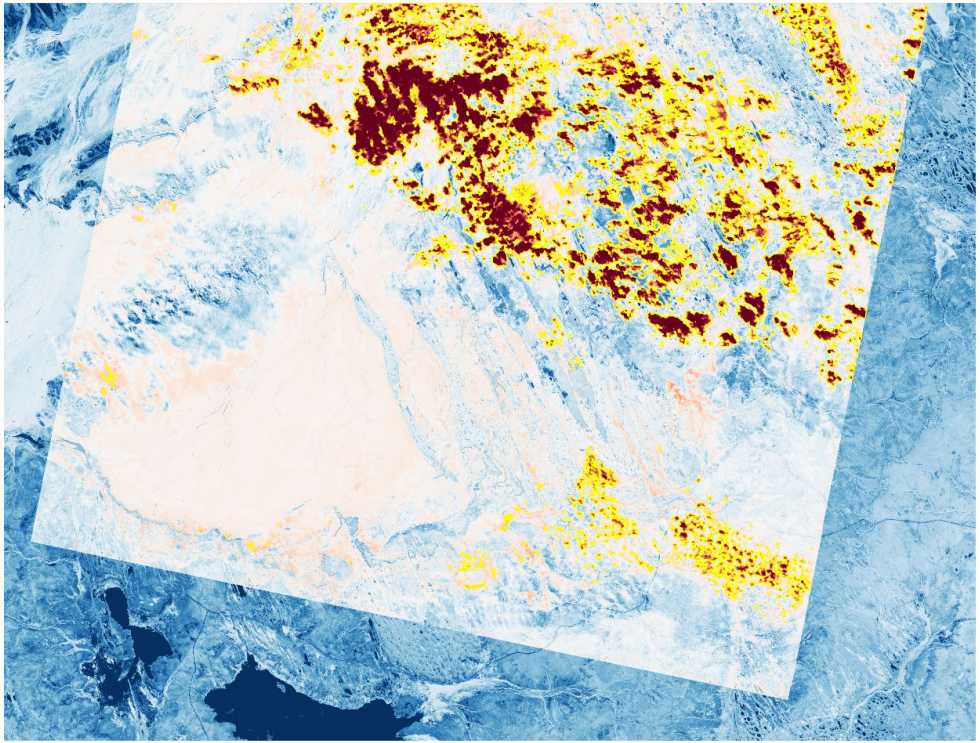


Cloud Score



First Run Cloud Mask

Final Result



Skills acquired during experience

QGIS

rasterio



Presentation skills

python™ jupyter ANACONDA®



TIME MANAGEMENT!

Communication

Remote internships are...interesting

- A lot of my initial challenges were just figuring out something that was entirely new to me.
- Many aspects of the experience was just about trying something new to see if it works, which was actually quite fun!

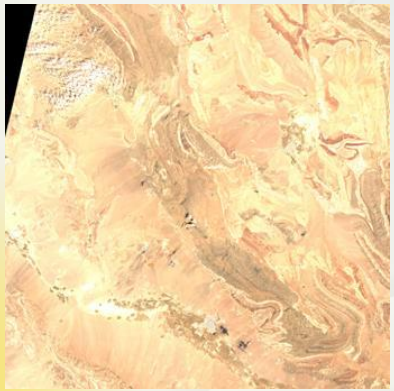
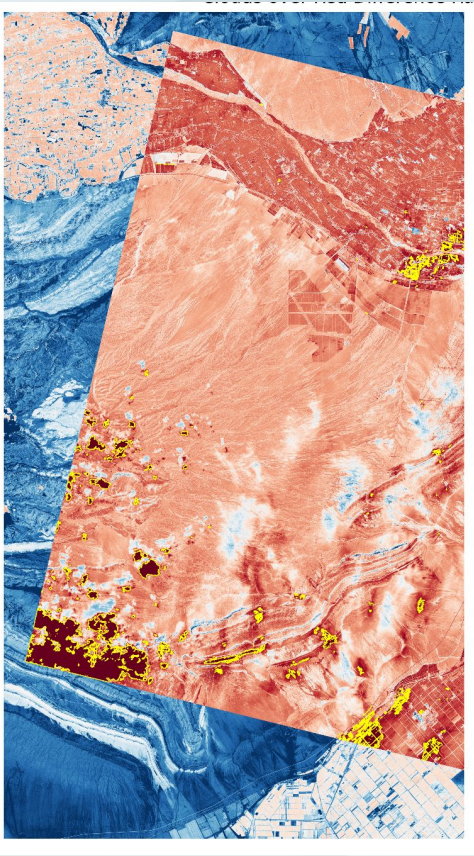




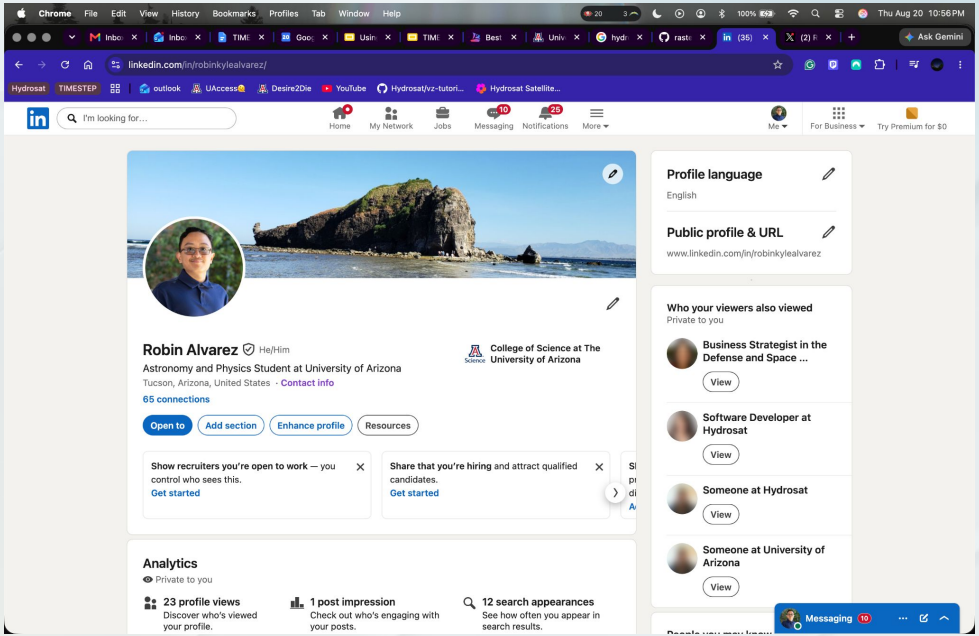
Image Credit:

ESA Sentinel-2, Copernicus



What did I learn about myself?

- The connections I made were awesome and extremely helpful.
- I was happy to hear that my contributions will improve Hydrosat's work in the future.
- I was apart of a company that aims for the betterment of humanity.



The Takeaway...

Grad School

My advice for future interns:

TRY SOMETHING NEW! You will see how much experience you can gain even if you were not expecting it.

Have an open mind. Be kind to yourself.

Thanks!





Do you have any questions?

robinalvarez@arizona.edu
+1 916 479 6540
linkedin.com/robinkylealvarez

CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon**, and infographics & images by **Freepik**

Acknowledgements: Thank you to my supervisor, Ian McGreer for guidance on this project. Tania Kleynhans for providing me VZ01 products for analysis, Nick Wilson for assistance in QGIS & Mission Tracker, and Neil Woodhouse for information and advice for cloud mask modeling. I also want to acknowledge the TIMESTEP team and my grad mentor, Jingyu Wang, for their invaluable guidance and providing me this opportunity to establish my career.